

# Financial Forecasting

## Recursion:

Recursion is a programming technique where a method calls itself repeatedly until a stopping condition (base case) is reached.

## Time Complexity of the Algorithm:

```
public static double forecast(double currentValue,
                             double growthRate,
                             int years) {

    if(years == 0) {
        return currentValue;
    }

    return forecast(
        currentValue * (1 + growthRate),
        growthRate,
        years - 1
    );
}
```

## Suppose:

years = 5

## Calls:

```
forecast(..., 5)
forecast(..., 4)
forecast(..., 3)
forecast(..., 2)
```

forecast(..., 1)  
forecast(..., 0)

Total calls = 6

For n years, approximately n recursive calls are made.

Time Complexity

$$T(n) = T(n-1) + O(1)$$

Therefore:

$$\text{Time Complexity} = O(n)$$

because the function executes once for every year.

### **How can it be optimized?**

The recursive solution can be optimized by:

1. Using an iterative loop, which reduces space complexity to **O(1)**.
2. Using the compound growth formula  $FV = PV(1 + r)^n$  with `Math.pow()`, which provides the most efficient solution with approximately **O(1)** time and **O(1)** space complexity.